

# Aditya Saini

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## EDUCATION

### Purdue University

West Lafayette, IN

*B.S. Computer Science (Systems Programming)*

*Graduation: May 2028 (Expected)*

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming, Multivariate Calculus

## EXPERIENCE

### Embedded Engineer

Sep. 2025 – Present

*Purdue Electric Racing*

*West Lafayette, IN*

- Developed modular RTOS-based vehicle firmware in C, integrating drivers and performing in-circuit debugging
- Implemented runtime telemetry and data acquisition to validate performance bottlenecks in production firmware
- Designed I2C and SPI HALs for STM32 platforms, standardizing high-frequency sensor communication
- Developed Nxtion LCD UART drivers and optimized command encoding for reliable real-time dashboard updates

### Controls Engineer

Dec 2025 – Present

*Purdue Space Program*

*West Lafayette, IN*

- Developed a flight dynamics simulation in MATLAB to model rigid body translational and rotational systems
- Implemented and tuned a PID controller for closed-loop stabilization of an unstable nonlinear control system
- Implemented embedded C drivers on ESP32 rocket test platform enabling real-time control and system validation

### Robotic Safety Research Assistant

January 2026 – July 2026

*PurSec, Purdue Computer Science*

*West Lafayette, IN*

- Developed robotic simulations using AI2THOR to find efficient jailbreaking methods for MLLM controlled robots
- Used AWS EC2 to run and gather results on experiments through EmbodiedBench in a headless environment
- Used Claude Code and Cursor to generate real world environment JSON files to perform robotic tasks

### Advanced Mathematics Teacher

June 2023 – July 2025

*Kumon North America*

*Edison, NJ*

- Developed tailored pedagogical strategies to explain complex mathematical concepts to a diverse student body, ensuring mastery of core principles for over 20+ concurrent students.
- Managed high-volume academic data entry and performance tracking, utilizing center-specific software to generate progress reports and identify learning bottlenecks for student development.

## PROJECTS

### Natural Language Shell | C, Python, OpenAI API, Git

June 2025 – January 2026

- Created a lightweight command-line shell in C supporting natural language commands with tokenizer, intent parsing, and bash mapping; integrated OpenAI API for complex commands/queries through OpenRouter.

### FRC Team 2554 | Java, OpenCV, Jetson Orin Nano, CUDA

September 2021 – August 2025

- Led the team to its best-ever season as Build President, bringing us to District Championships after 18+ years.
- Developed and led our team's first instance of Swerve Drive, tuned PID/FF loops, and developed auto commands to increase accuracy by 66% using custom vision pipelines to detect AprilTags for robot positioning.
- Managed finances and administrative requirements for the team in conjunction with robot design requirements.

### Random Tree Generator | Rust, CLI, Algorithms

April 2026

- Rust CLI that generates configurable general and binary trees with visual printing and built-in DFS traversals (preorder, inorder, postorder) for practicing recursive algorithms.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, SQL, JavaScript, HTML/CSS, R

**Embedded:** RTOS, STM32, ESP32, SPI, I2C, UART, CAN, TCP/IP, Arduino, Linux

**Developer Tools:** Git, Docker, Amazon Web Services, Claude Code, Cursor, Agent Orchestration

**Libraries:** pandas, NumPy, Matplotlib, React, Node.js, PyTorch, CUDA, OpenCV